The Notional Machine and Function-First Instruction

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The Molecular Sciences Software Institute

The Molecular Sciences Software Institute (the MolSSI)

- National Science Foundation Institute (launched 2016, renewed 2021)
- Collaborative effort by nine institutions in the United States
- Goal: Improve software, education, and training computational molecular science – a broad domain that includes quantum chemistry, computational materials science, and biomolecular simulation.

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Programming Curriculum at Cal Poly San Luis Obispo

- Goal: Enhance students understanding of chemistry by using programming to solve more complex physical chemistry problem
- Integrated Programming lessons in 5 physical chemistry courses
- 10+ years and 9 different instructors

Chemistry Learning Objective	Required programming skills
Use real equations of state	Write and use functions
Determine concentrations of reactants as a function of time for any rate equation	Solve single-variable and multi-variable differential equations numerically and plot the results.
Determine the energies of molecular orbitals	Create and diagonalize a matrix.

Are students learning transferable programming skills from the lessons in PChem?

- Some students consistently struggle with programming throughout the whole sequence.
- Some students are very resistant to learning programming.
- Skills very widely.
- Students self-report low confidence in their programming skills.

Multi-year study to assess students programming proficiency and confidence

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Major Lessons

- **Students will develop a notional machine about how programming works whether you explicitly teach them one or not.**
- Most students demonstrated level 2 programming abilities at the end of the course series.
- Most students appreciated that coding is a skill of growing importance.
- Most students identified at least some programming skills they thought were useful.
- 25% of students responding to the survey (n=68) said they had used their programing skills in another context, generally their undergraduate

What is the notional machine and why is it important?

- The notional machine is an abstract model of program execution; it's the students mental model for how they think program execution works
- The mental model is always executable; if you show the students something new, they "run" their mental model to decide what is going to happen

Some Misconceptions My Students Had

- If/and: students understood how an **and** expression worked inside an if conditional but not in other contexts
- Reactive variables: if a variable was changed they expected prior instances of that variable to also change
- Did not understand the scope of variables in functions
- Understand how to access matrix elements (**i.e. list[3]**) but not how to use a counter in a loop

Recommendations

Abstraction

Decomposition

Meta-Cognitive Awareness

Recommendations: Abstraction

- Explicitly teach students about a chosen notional machine
- Use multiple representations of programming concepts to promote abstraction and transfer learning.
- Relate abstractions used to solve programming problems to those used in Chemistry problems.

Recommendations: Decomposition

- Use sub-goals to divide programming problems into small steps
- Show examples of code and function reuse

Recommendations: Meta-Cognitive Awareness

- Teach an explicit programming problem solving process, such as the Design Recipe
- Relate iterative problem solving in programming to iterative problem solving in chemistry

The Design Recipe

Given a list of grades and an average class grade to meet, create a program that calculates the average, excluding the highest N grades and lowest N grades from the calculation, and returns whether curving is required or not. Curving is required if the calculated average is below a given minimum required average.

Figure 4.1: An example "black box" diagram
This diagram corresponds to Step 1 of our design recipe, Figure 4.3

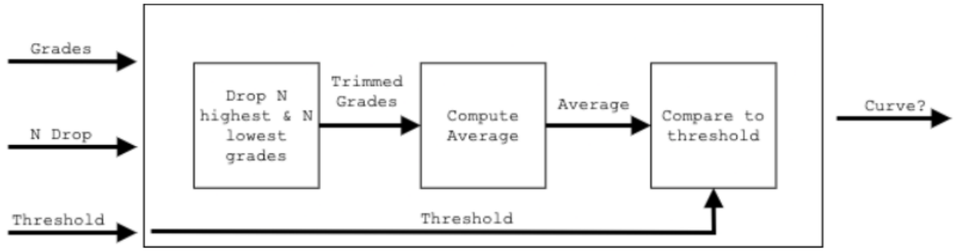

Figure 4.2: An example of sub-problems in a "black box" diagram
This diagram corresponds to Step 3 of our design recipe, Figure 4.3. In more complicated problems, it may be necessary to further decompose each "sub-problem black box" into a series of sub-sub-problem black boxes. This process can continue until the black boxes are small enough to easily be translated into pseudo-code.

Conclusions

- Students will develop a notional machine about how programming works whether you explicitly teach them one or not.
- To help students develop transferable programming skills:
  - Help students develop an abstract programming logic
  - Decompose problems into smaller parts
  - Plan and track their problem solving process.

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